JEE–MAIN EXAMINATION – JANUARY 2026

(HELD ON SATURDAY 24th JANUARY 2026) **TIME : 3:00 PM TO 6:00 PM**

MATHEMATICS

TEST PAPER WITH ANSWER

SECTION-A

1. Let $f(x)=\int\frac{7x^{10}+9x^8}{(1+x^2+2x^9)^2}dx$, $x > 0$, $\lim\limits_{x\to 0} f(x)=0$ and $f(1)=\frac{1}{4}$. If $A=\begin{bmatrix}0 & 0 & 1\\ \frac{1}{4} & f'(1) & 1\\ \alpha^2 & 4 & 1\end{bmatrix}$ and B = adj(adj A) be such that $|B| = 81$, then α^2 is equal to

(1) 2 (2) 3

(3) 1 (4) 4

2. Let the length of the latus rectum of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, (a > b), be 30. If its eccentricity is the maximum value of the function $f(t) = -\frac{3}{4}+2t-t^2$, then $(a^2 + b^2)$ is equal to -

(1) 516 (2) 256

(3) 496 (4) 276

3. Let the angles made with the positive x-axis by two straight lines drawn from the point P(2, 3) and meeting the line x + y = 6 at a distance $\sqrt{\frac{2}{3}}$ from the point P be θ_1 and θ_2. Then the value of $(\theta_1 + \theta_2)$ is :

(1) $\frac{\pi}{12}$ (2) $\frac{\pi}{6}$

(3) $\frac{\pi}{2}$ (4) $\frac{\pi}{3}$

4. Let $\vec{a}=2\hat{i}-\hat{j}-\hat{k}, \vec{b}=\hat{i}+3\hat{j}-\hat{k}$ and $\vec{c}=2\hat{i}+\hat{j}+3\hat{k}$. Let $\vec{v}$ be the vector in the plane of the vectors $\vec{a}$ and $\vec{b}$, such that the length of its projection on the vector $\vec{c}$ is $\frac{1}{\sqrt{14}}$. Then $|\vec{v}|$ is equal to

(1) $\frac{\sqrt{21}}{2}$ (2) 13

(3) $\frac{\sqrt{35}}{2}$ (4) 7

NTA (3)

5. Let $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be an A.P. of four terms such that each term of the A. P. and its common difference l are integers. If $\alpha_1+\alpha_2+\alpha_3+\alpha_4=48$ and $\alpha_1, \alpha_2, \alpha_3, \alpha_4 + l^4 = 361$ then the largest term of the A.P. is equal to

(1) 27 (2) 24 (3) 21 (4) 23

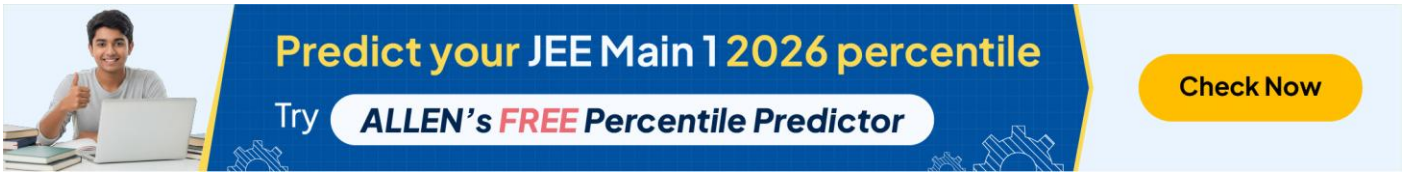
6. Let the image of parabola $x^2 = 4y$, in the line x – y = 1 be $(y + \alpha)^2 = b(x – c)$, a, b, c $\in \mathbb{N}$. Then a + b + c is equal to

(1) 12 (2) 4

(3) 6 (4) 8

7. $\left(\frac{1}{3}+\frac{4}{7}\right)+\left(\frac{1}{3^2}+\frac{1}{3}\times\frac{4}{7}+\frac{4^2}{7^2}\right)+\left(\frac{1}{3^3}+\frac{1}{3^2}\times\frac{4}{7}+\frac{1}{3}\times\frac{4^2}{7^2}+\frac{4^3}{7^3}\right)$ + ... upto infinite terms is equal to -

(1) $\frac{5}{2}$ (2) $\frac{7}{4}$ (3) $\frac{4}{3}$ (4) $\frac{6}{5}$



8. Let $P = [p_{ij}]$ and $Q = [q_{ij}]$ be two square matrices of order 3 such that $q_{ij} = 2^{(i+j-1)} p_{ij}$ and $\det(Q) = 2^{10}$. Then the value of $\det(\text{adj}(\text{adj } P))$ is :

(1) 32 (2) 16

(3) 81 (4) 124

9. The letters of the word "UDAYPUR" are written in all possible ways with or without meaning and these words are arranged as in a dictionary. The rank of the word "UDAYPUR" is :

(1) 1580 (2) 1578

(3) 1579 (4) 1581

10. The largest value of n, for which 40^n divides 60!, is

(1) 13 (2) 11

(3) 12 (4) 14

11. The sum of all values of α, for which the shortest distance between the lines $\frac{x+1}{\alpha} = \frac{y-2}{-1} = \frac{z-4}{-\alpha}$ and $\frac{x}{\alpha} = \frac{y-1}{2} = \frac{z-1}{2\alpha}$ is $\sqrt{2}$, is

(1) 8 (2) –6

(3) 6 (4) –8

12. Let $f(\alpha)$ denote the area of the region in the first quadrant bounded by $x = 0$, $x = 1$, $y^2 = x$ and $y = |\alpha x - 5| - |1 - \alpha x| + \alpha x^2$. Then $(f(0) + f(1))$ is equal to

(1) 9 (2) 14

(3) 7 (4) 12

13. If the domain of the function $f(x) = \sin^{-1}\left(\frac{1}{x^2 - 2x - 2}\right)$, is $(-\infty, \alpha] \cup [\beta, \gamma] \cup [\delta, \infty)$, then $\alpha + \beta + \gamma + \delta$ is equal to

(1) 2 (2) 4

(3) 3 (4) 5

14. Let $X = \{x \in \mathbb{N} : 1 \le x \le 19\}$ and for some $a, b \in \mathbb{R}$, $Y = \{ax + b : x \in X\}$. If the mean and variance of the elements of Y are 30 and 750, respectively, then the sum of all possible values of b is

(1) 20 (2) 80

(3) 100 (4) 60

15. Consider the following three statements for the function $f : (0, \infty) \to \mathbb{R}$ defined by

$f(x) = |\log_e x| - |x - 1|$:

(I) f is differentiable at all $x > 0$.

(II) f is increasing in (0, 1).

(III) f is decreasing in $(1, \infty)$.

Then.

(1) All (I), (II) and (III) are TRUE.

(2) Only (I) is TRUE.

(3) Only (II) and (III) are TRUE.

(4) Only (I) and (III) are TRUE.

16. Let y = y (x) be a differentiable function in the interval $(0, \infty)$ such that $y(1) = 2$.

and $\lim\limits_{t \to x}\left(\frac{t^2 y(x) - x^2 y(t)}{x - t}\right) = 3$ for each $x > 0$.

Then $2y(2)$ is equal to

(1) 18

(2) 23

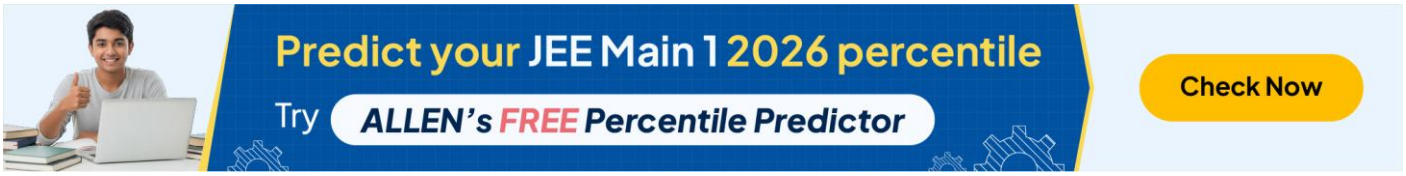
(3) 27

(4) 12

17. Let f be a function such that $3f(x) + 2f\left(\frac{m}{19x}\right) = 5x$, $x \neq 0$, where $m = \sum_{i=1}^{9} (i)^2$. Then $f(5) - f(2)$ is equal to

(1) –9 (2) 36

(3) 18 (4) 9



18. The smallest positive integral value of a, for which all the roots of $x^4 - ax^2 + 9 = 0$ are real and distinct, is equal to

(1) 9 (2) 3

(3) 4 (4) 7

19. Let $\vec{a} = 2\hat{i} - 5\hat{j} + 5k$ and $\vec{b} = \hat{i} - \hat{j} + 3k$. If $\vec{c}$ is a vector such that

$2(\vec{a} \times \vec{c}) + 3(\vec{b} \times \vec{c}) = \vec{0}$ and $(\vec{a} - \vec{b}).\vec{c} = -97$, then $|\vec{c} \times k|^2$ is equal to

(1) 193 (2) 233

(3) 218 (4) 205

20. Let [t] denote the greatest integer less than or equal to t. If the function

$$f(x) = \begin{cases} b^2 \sin\left(\frac{\pi}{2}\left[\frac{\pi}{2}(\cos x + \sin x)\cos x\right]\right) & , x < 0 \\ \frac{\sin x - \frac{1}{2}\sin 2x}{x^3} & , x > 0 \\ a & , x = 0 \end{cases}$$

is continuous at x = 0, then $a^2 + b^2$ is equal to

(1) $\frac{5}{8}$ (2) $\frac{9}{16}$

(3) $\frac{3}{4}$ (4) $\frac{1}{2}$

SECTION-B

21. If f(x) satisfies the relation $f(x) = e^x + \int_0^1 (y + xe^x) f(y)\, dy$, then e + f(0) is equal to _____.

22. Let (h, k) lie on the circle C : $x^2 + y^2 = 4$ and the point (2h + 1, 3k + 2) lie on an ellipse with eccentricity e. Then the value of $\frac{5}{e^2}$ is equal to _____.

23. Let $z = (1 + i)(1 + 2i)(1 + 3i)\ldots(1 + ni)$, where $i = \sqrt{-1}$. If $|z|^2 = 44200$, then n is equal to –

24. Let S be a set of 5 elements and P(S) denote the power set of S. Let E be an event of choosing an ordered pair (A, B) from the set P(S) × P(S) such that $A \cap B = \varnothing$. If the probability of the event E is $\frac{3^p}{2^q}$, where $p, q \in \mathbb{N}$, then p + q is equal to

25. The number of elements in the set

$\{x \in [0, 180^\circ] : \tan(x + 100^\circ) = \tan(x + 50^\circ)\tan x \tan(x - 50^\circ)\}$ is_____.



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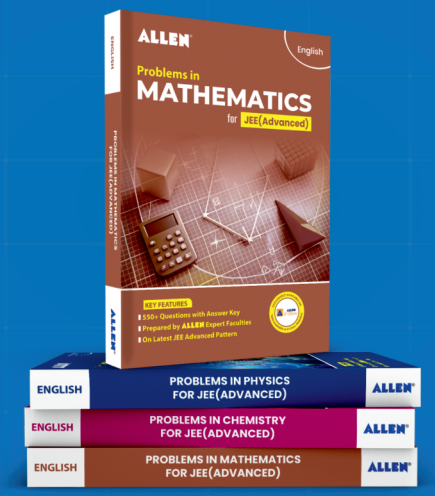


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